Target : E-Commerce Operations Analysis

(Name: Akhil Deshmukh, email: akdesh156@gmail.com)

Q1 - Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

A. Data type of all columns in the "customers" table.

Answer:

***Query*:**

```
SELECT column_name, data_type
FROM target.INFORMATION_SCHEMA.COLUMNS
WHERE table_name = 'customers';
```

Query Output:

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

***Insight*:** The customers table primarily consists of string data types for identification and geographical text, while zip codes are appropriately stored as integers.

***Recommendation*:** Since most columns are strings, they cannot be used for direct numerical calculations. Furthermore, when performing joins on customer_id and customer_unique_id, ensure these fields are properly indexed in BigQuery, as string-based joins are computationally heavier than integer joins.

B. Get the time range between which the orders were placed.

Answer:

***Query*:**

```
SELECT MIN(order_purchase_timestamp) AS first_order_date,
       MAX(order_purchase_timestamp) AS last_order_date
FROM target.orders;
```

Query Output:

Row	first_order_date	last_order_date
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

***Insight*:** The dataset has orders of around 2 years (25 months) starting from September 2016 to October 2018.

Recommendation: For better clarity we need to use datediff function to get exact days count between first and last order date, since 2016 and 2018 represent partial years of data, avoid using them for absolute year-over-year volume comparisons without normalizing the data first. Conducting a cohort analysis on the 2016 customers could yield valuable insights into user retention over this specific two-year span.

C. Count the Cities & States of customers who ordered during the given period.

Answer:

Query:

```
SELECT
  COUNT(DISTINCT c.customer_city) AS total_active_cities,
  COUNT(DISTINCT c.customer_state) AS total_active_states
FROM target.customers c
JOIN target.orders o ON c.customer_id = o.customer_id;
```

Query Output:

Row	total_active_cities	total_active_states
1	4119	27

Insight: Target has reached a high number of cities and is present in all the states.

Recommendation: Leverage this widespread active presence to negotiate favorable nationwide carrier rates rather than relying strictly on regional logistics providers. Continuously target newer, emerging cities within these states to deepen market penetration.

Q2 – In-depth Exploration:

A. Is there a growing trend in the no. of orders placed over the past years?

Answer:

Query:

```
SELECT
    EXTRACT(YEAR FROM order_purchase_timestamp) AS year,
    COUNT(order_id) AS total_orders
FROM target.orders
GROUP BY year
ORDER BY year;
```

Query Output:

Row	year	total_orders
1	2016	329
2	2017	45101
3	2018	54011

Insight: There is a distinct, explosive growth trend in the number of orders placed over the years. However, the initial jump between 2016 and 2017 appears drastically larger simply because the 2016 data only begins in September.

Recommendation: To accurately report the true growth rate to stakeholders, standardize the data to compare matching months (e.g., Jan-Oct) across all years. With steady year-over-year growth confirmed, the business must plan for proportional scaling in infrastructure, server capacity, and logistics networks.

B. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Answer:

Query:

```
SELECT
    EXTRACT(MONTH FROM order_purchase_timestamp) AS month,
    COUNT(order_id) AS total_orders
FROM target.orders
GROUP BY month
ORDER BY month;
```

Query Output:

Row	month	total_orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959
11	11	7544
12	12	5674

Insight: Volume exhibits clear seasonality, peaking during the mid-year months (May to August) and experiencing a sharp decline during September and October, with a slight recovery in November.

Recommendation: Capitalize on the peak mid-year months to launch new products and maximize marketing efforts. To smooth out the seasonal dip in September and October, deploy targeted campaigns and clearance sales to boost revenue.

C. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

Answer:

Query:

```
SELECT
    CASE
        WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
        WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN
'Morning'
        WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN
'Afternoon'
        ELSE 'Night'
    END AS time_of_day,
    COUNT(order_id) AS total_orders
FROM target.orders
GROUP BY time_of_day
ORDER BY total_orders DESC;
```

Query Output:

Row	time_of_day	total_orders
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

Insight: Shopping activity is heavily concentrated in the Afternoon (lunch breaks) and Night (post-work hours), making them the most popular times for order placement, while Dawn sees the least activity.

Recommendation: Schedule critical site maintenance and backend updates exclusively during the "Dawn" period to minimize revenue disruption. Conversely, trigger push notifications and execute flash deals during the high-traffic Afternoon and Night time slots to maximize conversions.

Q3 – Evolution of E-commerce orders in the Brazil region:

A. Get the month-on-month no. of orders placed in each state.

Answer:

Query:

```
SELECT c.customer_state, FORMAT_DATE('%Y-%m',DATE(o.order_purchase_timestamp)) AS
month,
    COUNT(o.order_id) AS orders
```

```
FROM target.orders o
JOIN target.customers c
ON o.customer_id = c.customer_id
GROUP BY 1, 2
ORDER BY 2, 3 DESC;
```

Query Output:

Row	customer_state	month	orders
1	SP	2016-09	2
2	RS	2016-09	1
3	RR	2016-09	1
4	SP	2016-10	113
5	RJ	2016-10	56
6	MG	2016-10	40
7	RS	2016-10	24
8	PR	2016-10	19
9	SC	2016-10	11
10	GO	2016-10	9

Insight: The state of São Paulo (SP) overwhelmingly dominates order volume month over month, frequently generating more orders than the next two highest states (RJ and MG) combined.

Recommendation: Treat 'SP' as the company's anchor market—ideal for testing new product launches and UX changes rapidly before rolling them out nationwide. Analyze the specific factors driving SP's dominance and apply those learnings to high-potential followers like RJ and MG, offering targeted incentives like free shipping.

B. How are the customers distributed across all the states?

Answer:

Query:

```
SELECT customer_state,
  COUNT(distinct customer_unique_id) AS user_count
FROM target.customers
GROUP BY 1
ORDER BY 2 DESC;
```

Query Output:

Row	customer_state	user_count
1	SP	40302
2	RJ	12384
3	MG	11259
4	RS	5277
5	PR	4882
6	SC	3534
7	BA	3277
8	DF	2075
9	ES	1964
10	GO	1952

Insight: The customer base is highly concentrated geographically, with the top 3 states (SP, RJ, MG) holding the vast majority of users, and SP alone holding more than double the users of its closest followers.

Recommendation: In these top 3 performing states, prioritize launching new products first to gather immediate feedback. To reduce reliance on a single geographic economic zone, launch geo-targeted acquisition campaigns and expand logistics infrastructure in lower-tier states to diversify the customer portfolio.

Q4 – Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

A. Get the % increase in the cost of orders from year 2017 to 2018 *(include months between Jan to Aug only)*.

You can use the "payment_value" column in the payments table to get the cost of orders.

Answer:

Query:

```
SELECT
    round(((SUM(CASE WHEN EXTRACT(YEAR from order_purchase_timestamp) = 2018 THEN
payment_value END) -
      SUM(CASE WHEN EXTRACT(YEAR from order_purchase_timestamp) = 2017 THEN
payment_value END)) /
      SUM(CASE WHEN EXTRACT(YEAR from order_purchase_timestamp) = 2017 THEN
payment_value END)) * 100, 2) AS pct_increase
FROM target.orders o JOIN target.payments p ON o.order_id = p.order_id
WHERE extract(month from order_purchase_timestamp) BETWEEN 1 AND 8
and o.order_status = 'delivered';
```

Query Output:

Row	pct_increase
1	143.33

Insight: Target achieved a massive 143.33% year-over-year spike in total payment value for identical Jan-Aug periods. This indicates successful market penetration, higher volume, and overall hyper-growth.

Recommendation: With top-line revenue more than doubling, the strategic priority must shift to retention and system stability. Reinvest a portion of this surging sales into loyalty programs to convert 2017/2018 first-time buyers into repeat customers. Plan for current server and logistics capacity immediately to ensure they can handle a similar growth wave in 2019.

B. Calculate the Total & Average value of order price for each state.

Answer:

Query:

```
SELECT
  c.customer_state,
  round(SUM(i.price),2) AS total_val,
  round(SUM(i.price) / COUNT(DISTINCT o.order_id), 2) AS avg_val
FROM target.customers c
JOIN target.orders o ON c.customer_id = o.customer_id
JOIN target.order_items i ON o.order_id = i.order_id
WHERE o.order_status = 'delivered'
GROUP BY 1
ORDER BY 2 DESC, 3 DESC;
```

Query Output:

Row	customer_state	total_val	avg_val
1	SP	5067633.16	125.12
2	RJ	1759651.13	142.48
3	MG	1552481.83	136.73
4	RS	728897.47	136.37
5	PR	666063.51	135.3
6	SC	507012.13	142.98
7	BA	493584.14	151.59
8	DF	296498.41	142.55
9	GO	282836.7	144.53
10	ES	268643.45	134.66

Insight: While the top 3 states drive the vast majority of total revenue, there is an inverse relationship in some regions where states with lower total volume actually exhibit a higher average order value, indicating a pattern to spend on higher-priced products.

Recommendation: Tailor merchandising strategies geographically: in states with higher average order values, focus marketing efforts on premium brands and high-end expensive products. Conversely, in states with lower averages but high volume, promote daily essentials and bulk discounts.

C. Calculate the Total & Average value of order freight for each state.

Answer:

Query:

```
SELECT
  c.customer_state,
  round(SUM(i.freight_value),2) AS total_freight,
  round(SUM(i.freight_value) / COUNT(DISTINCT o.order_id), 2) AS avg_freight
FROM target.customers c
JOIN target.orders o ON c.customer_id = o.customer_id
JOIN target.order_items i ON o.order_id = i.order_id
WHERE o.order_status = 'delivered'
GROUP BY 1
ORDER BY 2 DESC, 3 DESC;
```

Query Output:

Row	customer_state	total_freight	avg_freight
1	SP	702069.99	17.33
2	RJ	295750.44	23.95
3	MG	266409.84	23.46
4	RS	132575.32	24.8
5	PR	115645.29	23.49
6	BA	97553.67	29.96
7	SC	88115.65	24.85
8	PE	57082.56	35.83
9	GO	51375.65	26.25
10	DF	49624.94	23.86

Insight: States that generate the highest total freight costs (due to sheer volume) benefit from economies of scale, resulting in the lowest per-order average freight costs. Conversely, lower-volume states suffer from high average freight costs, making single orders less profitable.

Recommendation: Implement dynamic free-shipping thresholds. Introduce/update a higher cart minimum to qualify for free shipping in remote states with high average freight costs to protect profit margins, while offering lower cart minimums in states with low average freight costs to drive aggressive conversion.

Q5 – Analysis based on sales, freight and delivery time.

A. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.
You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- **time_to_deliver** = order_delivered_customer_date - order_purchase_timestam.
- **diff_estimated_delivery** = order_estimated_delivery_date - order_delivered_customer_date

Answer:

Query:

```
SELECT
    order_id,
    TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS
time_to_deliver,
    TIMESTAMP_DIFF(order_estimated_delivery_date, order_delivered_customer_date,DAY) AS
diff_estimated_delivery
FROM target.orders
WHERE order_status = 'delivered'
  AND order_delivered_customer_date IS NOT NULL
  AND order_estimated_delivery_date IS NOT NULL;
```

Query Output:

Row	order_id	time_to_deliver	diff_estimated_d...
1	bfbd0f9bdef84302105ad712db...	54	-36
2	98974b076b01553d49ee64679...	43	6
3	c4b41c36dd589e901f6879f25a...	36	14
4	d2292ff2201e74c5db154d1b7a...	29	20
5	95e01270fcbae986342340010...	30	19
6	ed8c7b1b3eb256c70ce0c7423...	44	5
7	5cc475c7c03290048eb2e742c...	68	-18
8	6b3ee7697a02619a0ace2b3f0a...	47	2
9	3b2ca3293a7ce539ea2379d70...	43	7
10	b2f92b2f7047cd8b35580d629d...	43	7

Insight: The variance (diff_estimated_delivery) reveals that the logistics estimation is highly unpredictable and inaccurate with either strong early delivery (around 2 weeks early) or strong late delivery (around 30 days late) . Positive numbers mean orders arrived early, but negative numbers indicate instances where carriers entirely failed to meet the promised window

Recommendation: Analyze the specific orders yielding negative values to identify poorly performing regional carriers and replace them. Simultaneously, tighten the prediction algorithm to be slightly more aggressive, as quoting overly long delivery times at checkout deters buyers.

B. Find out the top 5 states with the highest & lowest average freight value.

Answer:

Query:

```
WITH RankedStates AS (
  SELECT
    c.customer_state,
    ROUND(AVG(i.freight_value), 2) AS avg_freight,
    DENSE_RANK() OVER(ORDER BY AVG(i.freight_value) DESC) AS highest_rank,
    DENSE_RANK() OVER(ORDER BY AVG(i.freight_value) ASC) AS lowest_rank
  FROM target.customers c
  JOIN target.orders o ON c.customer_id = o.customer_id
  JOIN target.order_items i ON o.order_id = i.order_id
  where o.order_status = 'delivered'
  GROUP BY 1
)
SELECT
  CASE
    WHEN highest_rank <= 5 THEN 'Highest'
    WHEN lowest_rank <= 5 THEN 'Lowest'
  END AS category,
  customer_state,
  avg_freight AS average_freight_value
FROM RankedStates
WHERE highest_rank <= 5 OR lowest_rank <= 5
ORDER BY average_freight_value DESC;
```

Query Output:

Row	category	customer_state	average_freight_value
1	Highest	RR	43.09
2	Highest	PB	43.09
3	Highest	RO	41.33
4	Highest	AC	40.05
5	Highest	PI	39.12
6	Lowest	DF	21.07
7	Lowest	RJ	20.91
8	Lowest	MG	20.63
9	Lowest	PR	20.47
10	Lowest	SP	15.12

Insight: There is a striking disparity in average freight pricing across different states with the highest states averaging nearly triple the freight cost of the lowest states which heavily influences a customer's willingness to complete an order due to shipping cost sensitivity.

Recommendation: Explore a "Ship to Store" or centralized locker model in the top 5 highest-freight states. Consolidating individual deliveries into bulk shipments to a central pickup point will drastically lower these outlier freight averages. In states benefiting from low average

freight costs leverage the low logistics costs by running marketing campaigns highlighting "Fast & Cheap Delivery" as a core brand pillar.

C. Find out the top 5 states with the highest & lowest average delivery time.

Answer:

Query:

```
(SELECT 'Highest' as category, c.customer_state,
round(AVG(TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp,
DAY))) as average_delivery_time
FROM target.customers c
JOIN target.orders o ON c.customer_id = o.customer_id
WHERE order_status = 'delivered' and order_delivered_customer_date is not null
GROUP BY 2
ORDER BY 3 DESC LIMIT 5)

UNION ALL

(SELECT 'Lowest' as category, c.customer_state,
round(AVG(TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp,
DAY))) as average_delivery_time
FROM target.customers c
JOIN target.orders o ON c.customer_id = o.customer_id
WHERE order_status = 'delivered' and order_delivered_customer_date is not null
GROUP BY 2
ORDER BY 3 ASC LIMIT 5);
```

Query Output:

Row	category	customer_state	average_delivery_time
1	Highest	RR	29.0
2	Highest	AP	27.0
3	Highest	AM	26.0
4	Highest	AL	24.0
5	Highest	PA	23.0
6	Lowest	SP	8.0
7	Lowest	MG	12.0
8	Lowest	PR	12.0
9	Lowest	DF	13.0
10	Lowest	SC	14.0

Insight: The significant gap between the states with the lowest average delivery times (fastest deliveries) and those with the highest (slowest deliveries) highlights stark regional inequalities in the company's logistics infrastructure. The slowest states take roughly 3 to 4 times (nearly a month (26–29 days)) longer to receive goods compared to the fastest states (highly competitive 8 days).

Recommendation: Partner with localized final-mile logistics providers in the slowest regions to bypass larger, inefficient national carriers. Recalculate the max threshold for delivery time to

bring all states under a 15-day average. Achieving this requires injecting micro-fulfillment centers into the top 5 slowest states, as the current model is failing to provide a modern e-commerce experience to those regions. For the fastest states, proudly display "Fast Delivery" badges on product pages based on the user's IP address to boost checkout conversions.

D. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Answer:

Query:

```
SELECT c.customer_state, round(AVG(DATE_DIFF(order_estimated_delivery_date,
order_delivered_customer_date, DAY))) AS early_delivery_by_days
FROM target.orders o JOIN target.customers c ON o.customer_id = c.customer_id
WHERE order_status = 'delivered' and order_delivered_customer_date is not null and
order_estimated_delivery_date is not null
GROUP BY 1
ORDER BY 2 DESC LIMIT 5;
```

Query Output:

Row	customer_state	early_delivery_by_days
1	AC	20.0
2	AM	19.0
3	RO	19.0
4	AP	19.0
5	RR	16.0

Insight: States like AC, AM, and RO are receiving their packages a staggering 19 to 20 days earlier than the estimated date indicating that the current shipping algorithm is severely under-selling delivery speeds.

Recommendation: Immediately recalibrate the shipping estimate engine for these 5 states. By shortening the displayed delivery window on the checkout page, Target will capture buyers who might otherwise abandon their carts due to perceived slow shipping. Highlight "Faster than expected delivery" on the website for these regions to boost sales, and study their operational methods to replicate success elsewhere.

Q6 – Analysis based on the payments:

A. Find the month-on-month no. of orders placed using different payment types.

Answer:

Query:

```
SELECT p.payment_type,FORMAT_TIMESTAMP('%Y-%m', o.order_purchase_timestamp) AS month,
  COUNT(distinct o.order_id) AS monthly_orders
```

```
FROM target.orders o
JOIN target.payments p
  ON o.order_id = p.order_id
WHERE o.order_status = 'delivered'
GROUP BY payment_type,  month
ORDER BY month,  monthly_orders DESC;
```

Query Output:

Row	payment_type	month	monthly_orders
1	credit_card	2016-10	208
2	UPI	2016-10	51
3	voucher	2016-10	9
4	debit_card	2016-10	2
5	credit_card	2016-12	1
6	credit_card	2017-01	541
7	UPI	2017-01	188
8	voucher	2017-01	32
9	debit_card	2017-01	9
10	credit_card	2017-02	1249

Insight: Credit cards overwhelmingly dominate the payment landscape, showing consistent month-over-month growth and reliable usage, followed by UPI, while debit cards remain the least preferred method.

Recommendation: Leverage the massive volume of credit card transactions to negotiate lower processing fees with major providers. Provide exclusive offers (cashback, instant discounts) on credit cards to incentivize their use. For consistently low-performing payment methods, investigate underlying issues or consider removing them to streamline the checkout.

B. Find the no. of orders placed on the basis of the payment installments that have been paid.

Answer:

Query:

```
SELECT
  payment_installments,
  COUNT(DISTINCT p.order_id) AS total_orders
FROM target.payments p
JOIN target.orders o
  ON p.order_id = o.order_id
WHERE o.order_status = 'delivered'
GROUP BY payment_installments
ORDER BY total_orders DESC;
```

Query Output:

Row	payment_installments	total_orders
1	1	47586
2	2	12052
3	3	10147
4	4	6882
5	10	5137
6	5	5090
7	8	4122
8	6	3800
9	7	1560
10	9	618

Insight: The vast majority of completed transactions are paid in a single, upfront installment. However, there is a substantial secondary peak of users splitting payments between 2 to 10 installments, highlighting a strong reliance on financing for certain segments.

Recommendation: Ensure the single-payment option is heavily optimized and visually prominent at checkout. To boost conversion rates for mid-range purchases, offer targeted promotions like "0% interest for up to 6 months". Finally, map installment behavior to specific product categories so promotional financing is dynamically displayed for high-value items.

Executive Summary & Action Plan

Based on the comprehensive data analysis of Target's operations in Brazil, the following strategic themes and action items have emerged to guide operations in the upcoming fiscal year:

Geographic Concentration & Hyper-Growth: Target experienced a massive 143.33% year-over-year revenue spike. However, the customer base is highly consolidated in the Southeastern region (SP, RJ, MG).

- **Action**: Treat São Paulo as an "anchor market" for rapidly testing new products. To sustain this hyper-growth, deploy geo-targeted acquisition campaigns in central and northern states to diversify the customer portfolio, while aggressively scaling server and warehouse infrastructure.

Logistics Disparities: Delivery efficiency is highly unequal across Brazil. Remote states suffer from high freight costs and take up to 3x longer to receive orders, yet the estimation algorithm frequently predicts deliveries taking weeks longer than they actually do.

- **Action**: Recalibrate the delivery estimation engine to show faster, more accurate delivery windows at checkout to prevent cart abandonment. Transition to regional carrier partnerships in slow states, and implement dynamic cart minimums (higher minimums in high-freight regions) to protect profit margins.

Payment Customization: While upfront credit card payments dominate, a significant portion of the user base relies on 2–10-part installment plans to afford purchases.

- **Action**: Tie installment financing offers strictly to high-value product categories. Furthermore, launch targeted mid-day and evening promotional campaigns to align with peak purchasing hours (1:00 PM and 7:00 PM).

Data Quality & Engineering Caveats

- **Database Constraints:** The customers table utilizes strings for IDs. Engineering should index these fields to prevent slow query performance.

- **Logistics Tracking Errors:** A subset of orders returned negative delivery differentials (where the recorded delivery date was mathematically prior to the purchase timestamp). This indicates manual entry errors or time zone logging failures at the carrier level that must be sanitized for future automated reporting.
